

```
(venv1) C:\Users\kamil>pip install django
Collecting django
  Downloading Django-4.0.3-py3-none-any.whl (8.0 MB)
    | 8.0 MB 3.3 MB/s
Collecting tzdata
  Downloading tzdata-2022.1-py2.py3-none-any.whl (339 kB)
    | 339 kB 2.2 MB/s
Collecting asgiref<4,>=3.4.1
  Downloading asgiref-3.5.0-py3-none-any.whl (22 kB)
Collecting sqlparse>=0.2.2
  Using cached sqlparse-0.4.2-py3-none-any.whl (42 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
Successfully installed asgiref-3.5.0 django-4.0.3 sqlparse-0.4.2 tzdata-2022.1
```

Figure 3: Installation of Django

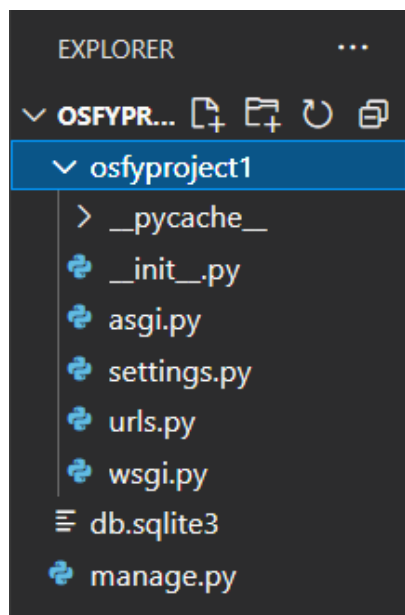


Figure 4: Contents of the project in VS Code

```
'osfyproject1.settings'
Starting development server at
http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

The `http://127.0.0.1:8000/` observed in the second last line of the above code is the base URL of all the operations we wish to perform. We will observe how we generate separate URLs as an extension of this for carrying out CRUD operations, in later sections.

Now let's head to VS Code and open our project by selecting it within the directory, the file path of which is `C:\Users\kamil\osfydir`.

Let us activate our virtual environment within the terminal of VS Code:

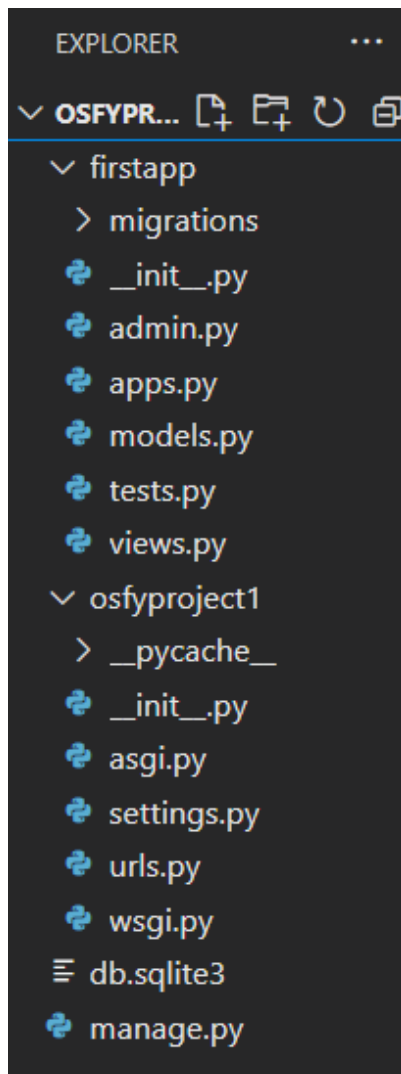


Figure 5: Updated file explorer in VS Code

```
Microsoft Windows [Version
10.0.19043.1586]
(c) Microsoft Corporation. All rights
reserved.
```

```
C:\Users\kamil\osfydir\
osfyproject1>workon venv1
(venv1) C:\Users\kamil\osfydir\
osfyproject1>
```

We now create an app within our project. Let's say that your 'project' is a website on the whole and the 'apps' are specific Web pages within the website. Since you wish to write the logics for APIs of different Web pages, you will want to write them in their respective `views.py` files. You will understand as we go further.

```
(venv1) C:\Users\kamil\osfydir\
osfyproject1>python manage.py startapp
firstapp
(venv1) C:\Users\kamil\osfydir\
osfyproject1>
```

Instead of getting an overview of the various purposes the different files within the app and the project serve, which might confuse you in the beginning, let's go about it step-by-step as we move along the path of obtaining the end goal of our task. First, let's create a table in our database with different kinds of fields. But hang on, we first need to establish a connection with the database in order to create a table within it. We will be using pgAdmin (PostgreSQL) for our task.

We first need to establish a connection between our project and the `osfyproject1` database. We need `psycopg2` to do so. Let's install `psycopg2` by entering a command within the prompt in VS Code. The following section of code shows the output of the installation operation.

```
(venv1) C:\Users\kamil\osfydir\
osfyproject1>pip install psycopg2
Collecting psycopg2
  Using cached psycopg2-2.9.3-cp310-
cp310-win_amd64.whl (1.2 MB)
Installing collected packages: psycopg2
Successfully installed psycopg2-2.9.3
WARNING: You are using pip version
```